

Melde's String - Determination of the Frequency of the Turing Fork

Aim:- To determine the frequency of an electrically maintained tuning fork.

Apparatus:

- The electrically maintained tuning fork, pulley, fine thread, metre scale, scalepan
- The arrangement consists of a electrically maintained tuning fork. One end of a fine string is attached to one of the prongs of the fork The other end of the string carries a scale pan which is passed over a small pulley

Principle:

According to the theory of vibration of strings, the fundamental frequency of vibration of a string of length l and mass per unit length m subjected to a tension T is given by

$$n = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

The string attached to the prong of the tuning fork can be arranged to vibrate in two ways. 1. Transverse mode of vibration, 2. longitudinal mode of vibration. If the prong of the fork vibrates in a direction perpendicular to the length of the string is called transverse mode of vibration. If the prong of the fork vibrates in a direction parallel to the length of the string is called longitudinal mode of vibration.

For the transverse mode of vibration, the frequency of vibration of the string will be the same as that of the fork.

$$\therefore \text{Frequency of the fork, } n = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

If M is the mass suspended at the end of the string, then $T = Mg$ where g is the acceleration due to gravity.

$$n = \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1}{2l} \sqrt{\frac{Mg}{m}} = \sqrt{\frac{g}{4m} \left(\frac{M}{l^2} \right)}$$

Here l is the length of the loop vibrating in fundamental mode. ie, l is the length of one loop of vibration.

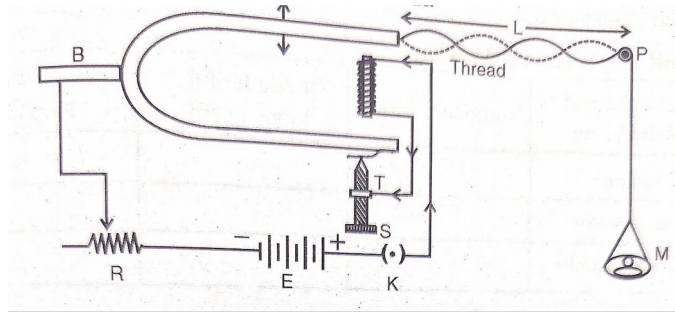
In the case of longitudinal mode of vibration, when fork makes one vibration the string makes only half vibration.

ie, Frequency of fork = 2xfrequency of the string

$$n = 2 \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1}{l} \sqrt{\frac{Mg}{m}} = \sqrt{\frac{g}{m} \left(\frac{M}{l^2} \right)}$$

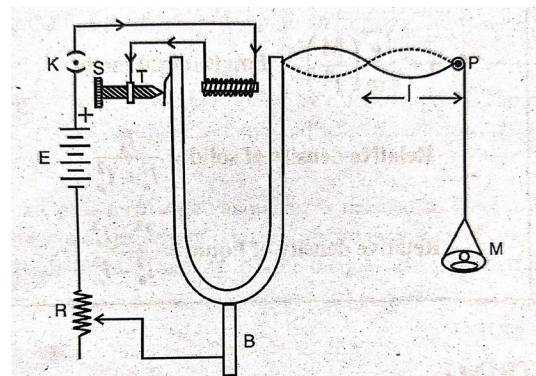
Procedure:

(1) Transverse mode:



- The electrical connections are made as shown in figure.
- Then the tuning fork is arranged in the transverse mode. For this the string is arranged horizontally with its length parallel to the prong of the fork. Then the fork vibrates in a direction perpendicular to the length of the string. This is the transverse mode.
- A mass M of about 2 or 3 grams is placed in the scale pan. The circuit is closed. The fork begins to vibrate.
- Now the fork is moved forward or back-ward slightly so that the length of the string can be adjusted to observe standing waves.
- Count the number of well defined loops as p. The length of the loops is measured by a metre scale as L. Therefore the length of one loop is L/p can be measured.
- The experiment is repeated for different values of mass and each time measure length of one loop L/p .
- Then calculate $M/(l^2)$ each time.
- M is the total mass suspended ie $M = \text{mass placed in the pan} + \text{mass of the scale pan}$.
- The frequency $n = \sqrt{\frac{g}{4m} \left(\frac{M}{l^2} \right)}$ can be calculated.
- To find m, take mass 10metre string and divide by 10 to get mass of one metre string.

(2) Longitudinal mode



- The tuning fork is arranged in longitudinal mode, for this the string is arranged horizontally with its length perpendicular to the prong of the fork. Then the fork vibrates in a direction parallel to the length of the string. This is the longitudinal mode of vibration.
- The experiment is repeated exactly as before for different masses and the mean value of $M/(l^2)$ is found out.

- The frequency of the vibrator is calculated as $n = \sqrt{\frac{g}{m} \left(\frac{M}{l^2} \right)}$

Observations:

Mass of 10 meter length of the wire =g

Mass per unit length of the wire = m =g=.....kg

(1) Transverse mode:

S.No	Mass in Scale pan	Total mass (including the mass of scale pan)	Number of loops p	Length of p loops in m	Length of one loop (l) in m	$\frac{M}{l^2}$ in Kg/m ²

$$\text{Mean } \frac{M}{l^2} = \text{.....Kg/m}^2$$

$$\text{Frequency of the vibrator } n = \sqrt{\frac{g}{4m} \left(\frac{M}{l^2} \right)} = \text{..... Hz}$$

(2) Longitudinal mode:

S.No	Mass in Scale pan	Total mass (including the mass of scale pan)	Number of loops p	Length of p loops in m	Length of one loop (l) in m	$\frac{M}{l^2}$ in Kg/m ²

$$\text{Mean } \frac{M}{l^2} = \text{.....Kg/m}^2$$

$$\text{Frequency of the vibrator } n = \sqrt{\frac{g}{m} \left(\frac{M}{l^2} \right)} = \text{..... Hz}$$

$$\text{Mean frequency } n = \text{.....Hz}$$

Results:

Frequency of the vibrator = Hz